THE DERIVED PUSHFORWARD COMMUTES WITH DERIVED LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the author additionally supplies an alternative proof in the case of enough points, and both are inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived pushforward along a morphism of ringed topoi and its computation by K-injective representatives. Same chapter, the construction of Rf_* and its computation using K-injective complexes. Supplies the functor whose commutation with derived limits is asserted and the representatives used in the proof.
- Derived limit R lim of an inverse system of complexes and its defining triangle. Same chapter together with the chapter Derived Categories, the construction of R lim through the product and the shift, and the associated distinguished triangle. Supplies the object R lim K_n and the triangle used in both proofs.
- Commutation of the underived pushforward with products and of restriction with limits. Same chapter, the behaviour of f_* with respect to products of sheaves and the exactness properties used in the alternate proof. Supplies the comparison making both proofs work.

Source reference. Original lemma 0941: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5067–5067. Navigation pointer invoked at proof line 5174; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 07D9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5119–5119. Navigation pointer invoked at proof line 5119; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07A6: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3990–3990. Navigation pointer invoked at proof line 5192; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0038: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5194–5194. Navigation pointer invoked at proof line 5194; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. Then Rf_* commutes with $R\lim$, i.e., Rf_* commutes with derived limits.*

Proof. Let (K_n) be an inverse system of objects of $D(\mathcal{O})$. By induction on n we may choose actual complexes $\mathcal{K}_n^\bullet$ of $\mathcal{O}$-modules and maps of complexes $\mathcal{K}_{n+1}^\bullet \to \mathcal{K}_n^\bullet$ representing the maps $K_{n+1} \to K_n$ in $D(\mathcal{O})$. In other words, there exists an object K in $D(\mathcal{C} \times \mathbf{N})$ whose associated inverse system is the given one. Next, consider the commutative diagram

$$\begin{array}{ccc} Sh(\mathcal{C} \times \mathbf{N}) & \xrightarrow{g} & Sh(\mathcal{C}) \\ {\scriptstyle f\times 1}\downarrow & & \downarrow{\scriptstyle f} \\ Sh(\mathcal{C}' \times \mathbf{N}) & \xrightarrow{g'} & Sh(\mathcal{C}') \end{array}$$

of morphisms of topoi. It follows that $R\lim R(f \times 1)_* K = Rf_* R\lim K$. Working through the definitions and using Lemma 0941 we obtain that $R\lim(Rf_* K_n) = Rf_*(R\lim K_n)$.

Alternate proof in case $\mathcal{C}$ has enough points. Consider the defining distinguished triangle

$$R\lim K_n \to \prod K_n \to \prod K_n$$

in $D(\mathcal{O})$. Applying the exact functor Rf_* we obtain the distinguished triangle

$$Rf_*(R\lim K_n) \to Rf_*\left(\prod K_n\right) \to Rf_*\left(\prod K_n\right)$$

in $D(\mathcal{O}')$. Thus we see that it suffices to prove that Rf_* commutes with products in the derived category (which are not just given by products of complexes, see Injectives, Lemma 07D9). However, since Rf_* is a right adjoint by Lemma 07A6 this follows formally (see Categories, Lemma 0038). Caution: Note that we cannot apply Categories, Lemma 0038 directly as $R\lim K_n$ is not a limit in $D(\mathcal{O})$. □